

radio waves range is:

- ☐ a. all frequencies below VHF
- ☐ b. UHF up to EHF
- ☒ c. VLF up to 1GHz
- ☐ d. VHF up to UHF

Radio waves, for the most part, are:

- ☐ a. multidirectional
- ☒ b. omnidirectional.
- ☐ c. unidirectional.
- ☐ d. bidirectional.

A light signal is travelling through a fiber. What is the delay in the signal if the length of fiber optic cable is 100m (assume a propagation speed is 2×10^8):

- ☐ a. 0.5 ms
- ☐ b. 0.05 ms
- ☐ c. 5 ms
- ☒ d. 0.5 μ s

Refraction of light occurred when:

- ☐ a. angle of incidence equal to reflection angle
- ☒ b. angle of incidence greater than critical angle
- ☐ c. angle of incidence less than or equal to critical angle
- ☐ d. angle of incidence equal to critical angle

The satellite communication is:

- ☐ a. terrestrial line of sight communication
- ☒ b. microwave line of sight communication
- ☐ c. optical fiber communication
- ☐ d. sky wave communication

Data rate is maximum by :

- ☒ a. optical fibers.
- ☐ b. coaxial cable
- ☐ c. free space
- ☐ d. twisted-pair

The following portion of the electromagnetic spectrum is not used in telecommunication:

- ☐ a. infrared light
- ☐ b. Microwave
- ☒ c. cosmic rays
- ☐ d. ocean waves

The bandwidth of the light for the wavelength range (1000-1200) nm is ($c = 2 \times 10^8$):

- ☒ a. 33 THz
- ☐ b. 33 GHz
- ☐ c. 33 MHz
- ☐ d. 33 kHz